ELECTRONIC RETAIL SALES DASHBOARD USING POWER BI

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INTRODUCTION

The Electronic Retail Sales Dashboard is a business intelligence and data visualization project developed using Microsoft Power BI to analyse the sales performance of an electronic retail business. Electronic retail businesses deal with different product categories, multiple store locations, customer segments, and large volumes of sales transactions. Analysing these details manually can be time-consuming and may make it difficult to understand overall business performance.

The main purpose of this dashboard is to transform raw retail sales data into meaningful visual information that helps users monitor revenue, profit, quantity sold, and order performance. It provides an overview of the business through KPI cards, charts, slicers, and detailed tables. The dashboard allows users to compare revenue across product categories, identify cities generating higher sales, examine profit distribution, and understand revenue contributions from different store types.

The dashboard also focuses on monthly sales trends and individual store performance. Interactive slicers are provided to filter information based on year, store type, state, customer segment, and product category. These filters allow users to analyse the data according to specific business requirements.

An additional KPI tooltip page is also developed to display important performance indicators, including Total Revenue, Total Profit, Total Orders, and Average Order Value. This feature provides a quick summary of important business information and improves the interactive experience of the dashboard.

Overall, the project demonstrates how Power BI can be used to analyse electronic retail sales data, identify business trends, compare performance across different categories, and support effective business decision-making.

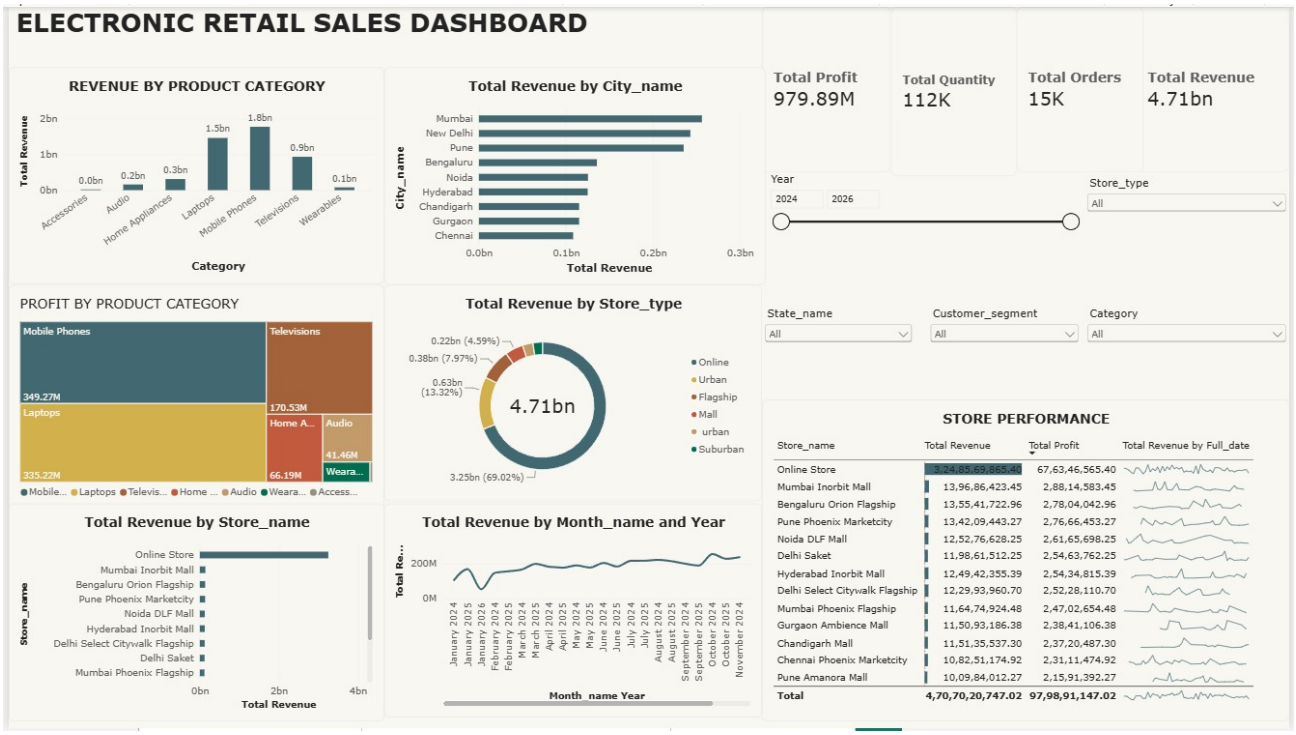
OBJECTIVES OF THE PROJECT

The main objectives of developing the Electronic Retail Sales Dashboard are:

1. To analyse the overall sales performance of an electronic retail business.

2. To calculate and monitor important business indicators such as Total Revenue, Total Profit, Total Quantity, and Total Orders.
3. To compare revenue generated by different electronic product categories.
4. To identify cities contributing the highest revenue to the business.
5. To analyse profit distribution across various product categories.
6. To understand the revenue contribution of different store types.
7. To compare the sales performance of individual stores.
8. To examine monthly revenue trends across different years.
9. To provide detailed store-level information through an interactive performance table.
10. To implement slicers for filtering sales data based on year, store type, state, customer segment, and category.
11. To create a KPI tooltip page that displays additional business performance indicators.
12. To transform retail sales data into meaningful insights that support business planning and decision-making.

DATASET DESCRIPTION



The dataset used in this project contains electronic retail sales information related to products, customers, stores, orders, and transactions. It includes different fields required to analyse the revenue and profitability of an electronic retail business.

The dataset contains information such as product category, city name, state name, store name, store type, customer segment, order quantity, sales revenue, profit, order details, and transaction dates.

The product category field is used to compare sales across categories such as Accessories, Audio, Home Appliances, Laptops, Mobile Phones, Televisions, and Wearables. The city and store-related fields help analyse geographical sales distribution and individual store performance.

The date-related fields, including Month Name and Year, are used to examine revenue trends over time. Customer Segment and Store Type fields support comparisons between different customer groups and retail channels.

Power BI is used to organize, transform, and summarize the dataset. Appropriate calculations and aggregations are applied to generate KPIs and visualizations. The final dashboard presents the information in an interactive format, making it easier to understand business performance from different perspectives.

DASHBOARD COMPONENTS AND KPI CARDS

Key Performance Indicators (KPIs) are used to display important business information in a summarized format. They help users understand the overall performance of the electronic retail business without having to examine every chart individually.

The main dashboard contains four KPI cards.

1. Total Profit



The Total Profit KPI displays approximately 979.89 million.

It represents the total profit generated from the sales activities included in the dataset. Profit is an important indicator because it shows the financial gain remaining after considering the costs included in the dataset's profit calculation.

Monitoring total profit helps the business understand its overall profitability. It also allows users to compare profit across product categories, stores, and other business segments to identify areas contributing more to the company's earnings.

2. Total Quantity



The Total Quantity KPI displays approximately 112K units.

It represents the total quantity of electronic products sold during the selected period. This indicator helps measure the sales volume of the business.

By monitoring total quantity, users can understand the overall demand for products and compare the number of units sold across different categories. It is useful for identifying popular products and supporting inventory and stock-planning decisions.

3. **Total Orders**

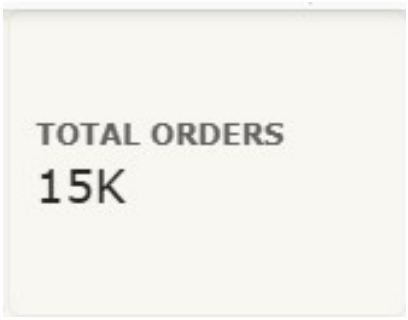


The Total Orders KPI displays approximately 15K orders.

It represents the total number of orders recorded in the dataset. This indicator provides an overview of customer purchasing activity.

Monitoring the total number of orders helps the business understand its sales activity and compare order performance across different periods, stores, and customer segments.

4. **Total Revenue**



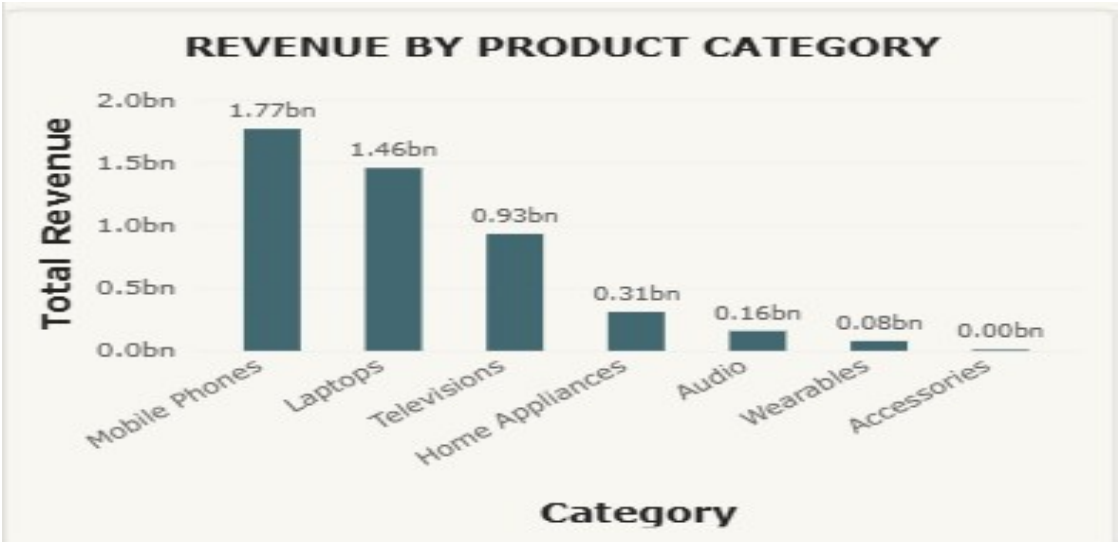
The Total Revenue KPI displays approximately 4.71 billion.

It represents the total sales revenue generated by the electronic retail business. Revenue is one of the most important indicators for evaluating business performance.

This KPI helps users understand the overall sales value generated during the selected period. It can also be compared with profit to understand the relationship between sales generation and profitability.

5. Dashboard Charts and Visualizations

1. Revenue by Product Category



A column chart is used to compare the total revenue generated by different electronic product categories.

The horizontal axis represents the product categories, while the vertical axis represents total revenue. The categories displayed are:

- Accessories
- Audio
- Home Appliances
- Laptops
- Mobile Phones
- Televisions
- Wearables

The chart shows that Mobile Phones generate the highest revenue, at approximately 1.8 billion, followed by Laptops at around 1.5 billion. Televisions contribute approximately 0.9 billion, while Home Appliances contribute around 0.3 billion.

Audio and Wearables contribute comparatively lower revenue, while Accessories generate the smallest amount among the displayed categories.

This visualization helps the business identify its major revenue-generating product categories. It also supports product-level business planning by showing which categories contribute more to overall sales.

2. Total Revenue by City Name



A horizontal bar chart is used to compare the total revenue generated across different cities.

The vertical axis represents city names, while the horizontal axis represents total revenue.

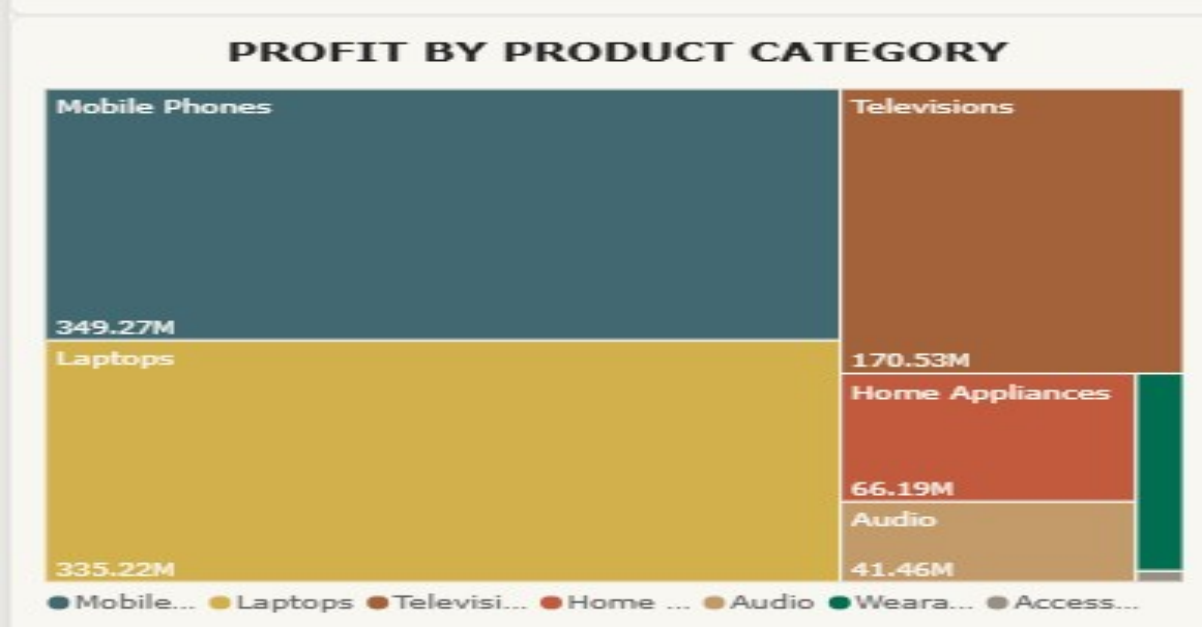
The cities displayed include Mumbai, New Delhi, Pune, Bengaluru, Noida, Hyderabad, Chandigarh, Gurgaon, and Chennai.

The chart shows that Mumbai generates the highest revenue, followed by New Delhi and Pune. Bengaluru also contributes a considerable amount of revenue, while Chennai and some of the other listed cities generate comparatively lower revenue.

This visualization helps the business understand its geographical sales performance. It allows users to identify high-performing cities and compare revenue contributions across different locations.

The information can support decisions related to store expansion, marketing strategies, and regional sales planning.

3. Profit by Product Category



A tree map chart is used to display the distribution of total profit across different product categories.

Each rectangular section represents a product category, and the size of the rectangle indicates its relative contribution to total profit. The chart also displays profit values within the categories.

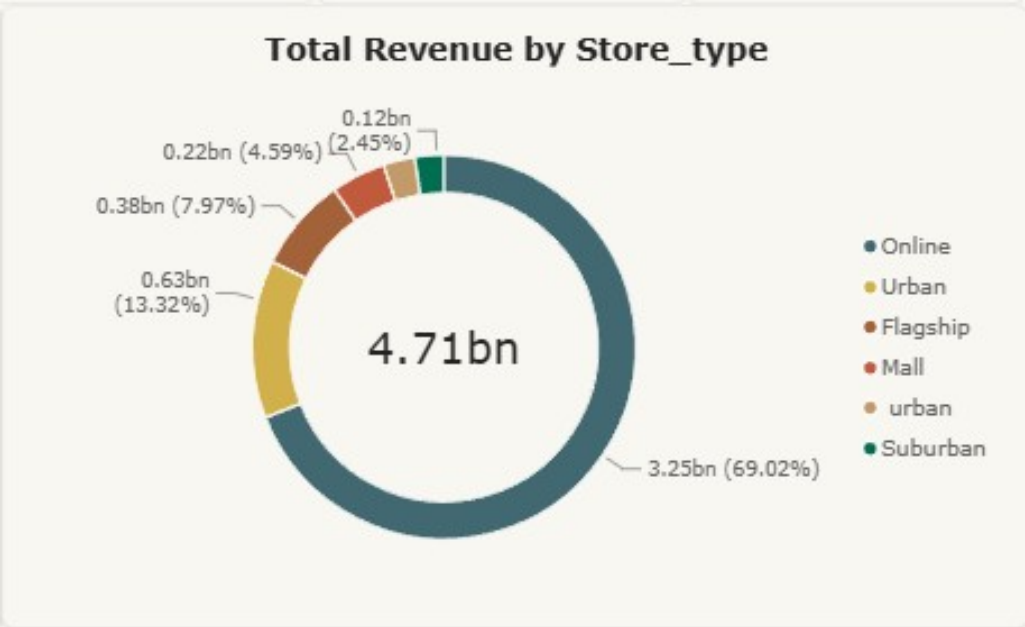
The visualization shows that Mobile Phones contribute approximately 349.27 million in profit, followed by Laptops at around 335.22 million. Televisions contribute approximately 170.53 million, while Home Appliances contribute around 66.19 million.

Audio contributes approximately 41.46 million, while Wearables and Accessories represent smaller portions of the displayed profit distribution.

This chart helps users understand which product categories contribute more to business profitability. It is particularly useful for comparing profit contributions because the relative sizes of the rectangles make the differences easier to identify.

The business can use this information to review product performance and plan its sales and pricing strategies.

5.Total Revenue by Store Type



A doughnut chart is used to analyse the total revenue contribution of different store types.

The chart displays the following store categories:

- Online
- Urban
- Flagship
- Mall
- urban
- Suburban

The visualization shows that Online stores contribute approximately 3.25 billion, representing around 69.02% of total revenue. Urban stores contribute approximately 0.63 billion or 13.32%, while Flagship stores contribute around 0.38 billion or 7.97%.

Mall stores contribute approximately 0.22 billion or 4.59%, with the remaining categories contributing smaller portions.

This visualization helps users understand how different store types contribute to overall revenue. The chart indicates that Online stores account for the largest share of sales revenue.

This information can help the business evaluate the performance of its sales channels and understand the importance of online retail operations.

6.Total Revenue by Store Name



A horizontal bar chart is used to compare revenue generated by individual stores.

The vertical axis represents store names, while the horizontal axis represents total revenue.

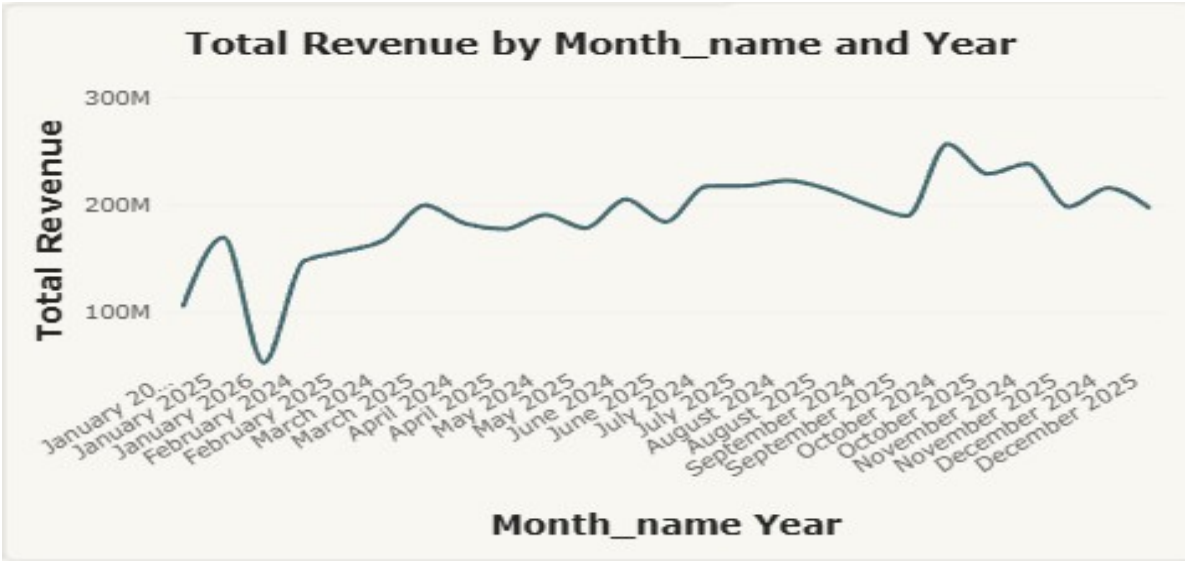
The chart includes stores such as Online Store, Mumbai Inorbit Mall, Bengaluru Orion Flagship, Pune Phoenix Marketcity, Noida DLF Mall, Hyderabad Inorbit Mall, and other retail locations.

The Online Store generates the highest revenue among the displayed stores, while the remaining physical stores contribute comparatively smaller amounts.

This visualization helps the business identify its highest-performing stores and compare revenue across different retail locations.

It is useful for evaluating individual store performance and identifying locations that may require additional marketing or operational attention.

5. **Total Revenue by Month Name and Year**



A line chart is used to analyse monthly revenue trends across different years.

The horizontal axis represents the Month Name and Year, while the vertical axis represents total revenue.

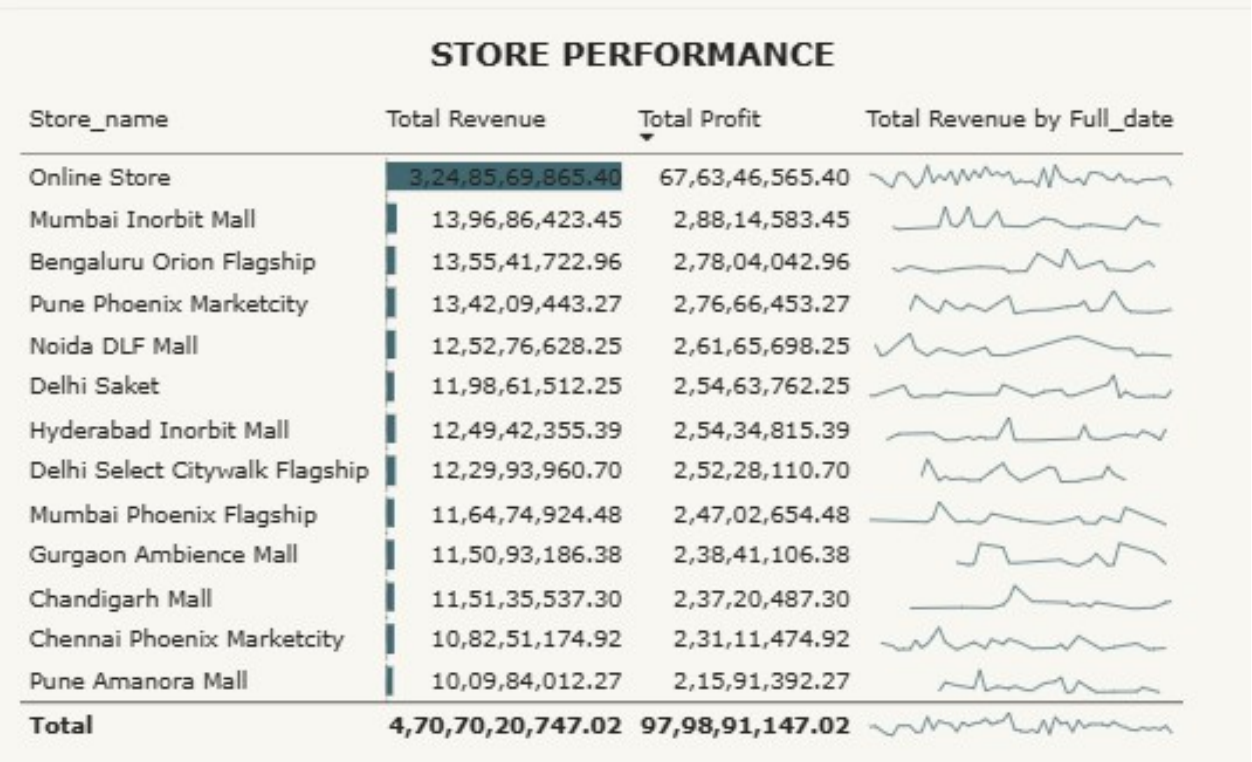
The chart displays revenue movement across the available months and helps users understand how sales performance changes over time.

The visualization shows variations in monthly revenue, including periods of increases and decreases. It also helps identify months in which revenue remains relatively stable or reaches higher levels.

This chart is useful for understanding seasonal sales patterns, comparing performance across different periods, and identifying changes in revenue over time.

The business can use these observations for sales forecasting, promotional planning, and inventory management.

8.Store Performance Table



STORE PERFORMANCE

Store_name	Total Revenue	Total Profit	Total Revenue by Full_date
Online Store	3,24,85,69,865.40	67,63,46,565.40	
Mumbai Inorbit Mall	13,96,86,423.45	2,88,14,583.45	
Bengaluru Orion Flagship	13,55,41,722.96	2,78,04,042.96	
Pune Phoenix Marketcity	13,42,09,443.27	2,76,66,453.27	
Noida DLF Mall	12,52,76,628.25	2,61,65,698.25	
Delhi Saket	11,98,61,512.25	2,54,63,762.25	
Hyderabad Inorbit Mall	12,49,42,355.39	2,54,34,815.39	
Delhi Select Citywalk Flagship	12,29,93,960.70	2,52,28,110.70	
Mumbai Phoenix Flagship	11,64,74,924.48	2,47,02,654.48	
Gurgaon Ambience Mall	11,50,93,186.38	2,38,41,106.38	
Chandigarh Mall	11,51,35,537.30	2,37,20,487.30	
Chennai Phoenix Marketcity	10,82,51,174.92	2,31,11,474.92	
Pune Amanora Mall	10,09,84,012.27	2,15,91,392.27	
Total	**4,70,70,20,747.02**	**97,98,91,147.02**	

A detailed table is included to provide store-level sales and profitability information.

The table contains the following columns:

- Store Name
- Total Revenue
- Total Profit
- Total Revenue by Full Date

The Store Name column identifies individual retail locations. Total Revenue displays the sales value generated by each store, while Total Profit displays the profit contributed by the corresponding store.

The Total Revenue by Full Date column contains small trend charts, known as sparklines, which show changes in revenue over time for each store.

Sparklines provide a compact visual representation of performance without requiring a separate chart for every store.

The table also includes an overall total row, which displays approximately 4.71 billion in revenue and 979.89 million in profit.

This visualization helps users compare individual store performance, understand revenue and profit contributions, and examine sales trends over time.

6. SLICERS AND FILTERS

Slicers are interactive filtering elements in Power BI that allow users to display information based on selected conditions. The dashboard contains five major filtering options.

1. **Year Slicer**



The Year slicer is used to filter sales information based on the selected year.

The dashboard displays a year range from 2024 to 2026 through a slider. Users can adjust the slider to select a particular year or a range of years.

This slicer helps users compare sales performance across different periods. It also allows users to examine how revenue, profit, and order activity change over time.

2.Store Type Slicer



The Store Type slicer is used to filter the dashboard based on different store categories.

Users can select a specific store type, such as Online, Urban, Flagship, Mall, or Suburban, depending on the available dataset values.

This filter helps users analyse the performance of individual retail channels and understand how different store types contribute to total revenue and profit.

2. **State Name Slicer**

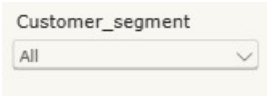


The State Name slicer allows users to filter sales information based on the state in which the stores are located.

By selecting a particular state, users can examine the sales and profitability of stores within that geographical area.

This filter is useful for regional performance analysis and helps the business compare sales contributions across different states.

4. Customer Segment Slicer



The Customer Segment slicer is used to filter information according to different customer groups.

It allows users to focus on the purchasing activity and revenue generated by a selected customer segment.

This feature helps the business understand customer-related sales patterns and compare the contribution of different customer groups.

5. Category Slicer



The Category slicer allows users to filter the dashboard based on electronic product categories.

Users can select categories such as Mobile Phones, Laptops, Televisions, Audio, Home Appliances, Wearables, and Accessories.

This slicer helps users examine the revenue and profit contribution of a specific product category across cities, stores, and different time periods.

Importance of Slicers

All five slicers improve the interactivity of the dashboard. Users can apply one or more filters to analyse specific portions of the dataset.

For example, users can select a particular year, state, store type, and product category to understand how that combination affects revenue and profit.

This makes the dashboard more flexible and allows users to perform focused business analysis without creating separate reports for every category.

7. KPI TOOLTIP PAGE



A separate KPI tooltip page is developed to display additional business performance indicators. It contains four KPI cards arranged in a two-by-two layout.

The tooltip page displays the following indicators:

1. **Total Revenue**

The Total Revenue indicator displays approximately 4.71 billion.

It represents the overall sales revenue generated by the business and provides a quick understanding of the financial value of sales.

2. Total Profit

The Total Profit indicator displays approximately 979.89 million.

It represents the overall profit generated through the sales activities. It helps users understand the profitability of the business.

3. Total Orders

The Total Orders indicator displays approximately 15K.

It represents the total number of orders recorded in the dataset and provides an overview of customer purchasing activity.

4. Average Order Value

The Average Order Value (AOV) indicator displays approximately 313.80K.

Average Order Value represents the average revenue generated per order. It helps the business understand how much revenue is generated, on average, from each order.

The general formula is:

Average Order Value = Total Revenue / Total Orders

This KPI is useful for understanding customer spending patterns and evaluating the value of individual orders.

Purpose of the Tooltip Page

The tooltip page provides a compact summary of important business indicators in a separate report page. When configured as a report page tooltip and assigned to the relevant visual, it can display these KPIs when users hover over a data point.

This feature allows users to access additional performance information without navigating away from the main dashboard.

It also helps keep the main dashboard organized while providing a more detailed interactive experience.

8. Dashboard Interactivity and Features

The Electronic Retail Sales Dashboard includes interactive elements that allow users to explore sales information from different perspectives.

The slicers provide filtering based on year, store type, state, customer segment, and product category. The charts and table can respond to these selections according to their configured visual interactions.

The KPI tooltip page provides additional information through hover-based interaction when it is assigned to the appropriate visual.

The Store Performance table also includes sparklines, which provide a quick view of revenue movement over time for individual stores.

These features improve the usability of the dashboard by allowing users to examine overall business performance as well as specific sales categories and store-level information.

9. Insights from the Dashboard

The Electronic Retail Sales Dashboard provides several important business observations:

1. The business generates approximately 4.71 billion in total revenue and 979.89 million in total profit.
2. Approximately 112K product units have been sold, with around 15K orders recorded in the dataset.
3. Mobile Phones generate the highest revenue among the displayed product categories, followed by Laptops.
4. Mobile Phones also contribute the highest profit, followed by Laptops, indicating their significant contribution to both revenue and profitability.
5. Mumbai generates the highest revenue among the displayed cities, followed by New Delhi and Pune.
6. Online stores contribute approximately 69.02% of total revenue, making them the largest revenue-generating store type.
7. The Online Store has the highest revenue among the individual stores displayed in the chart.
8. The monthly revenue line chart shows variations across the available months, indicating changes in sales performance over time.
9. The Store Performance table allows users to compare the revenue and profit of individual stores and examine their revenue trends using sparklines.
10. The KPI tooltip page provides additional information, including Average Order Value, which helps users understand the average revenue generated per order.

These observations help identify the major revenue-generating products, high-performing cities, important sales channels, and individual store contributions.

10. Importance of the Dashboard in Retail Business Analysis

The Electronic Retail Sales Dashboard demonstrates how business intelligence can support retail management and sales analysis.

Retail businesses need to monitor sales performance across products, locations, customers, and store channels. By combining these details in a single dashboard, Power BI makes it easier to identify important business patterns and compare performance.

The dashboard helps businesses understand which product categories generate higher revenue, which stores perform better, and which geographical areas contribute more to sales. It also provides information about profitability, order volume, and monthly revenue trends.

The use of slicers allows users to perform focused analysis, while the KPI tooltip page provides additional information without overcrowding the main dashboard.

These features can support business planning, sales strategy development, inventory management, regional performance evaluation, and customer analysis.

11. Conclusion

The Electronic Retail Sales Dashboard demonstrates the practical application of Microsoft Power BI in analysing and visualizing electronic retail business data. The dashboard provides a comprehensive overview of revenue, profit, quantity sold, orders, product categories, cities, store types, and individual store performance.

The different visualizations serve specific analytical purposes. Column charts compare revenue across product categories, horizontal bar charts identify high-performing cities and stores, tree maps represent profit distribution, doughnut charts display store-type revenue contributions, and line charts analyse monthly revenue trends.

The Store Performance table provides detailed information about revenue and profit at the individual store level, while sparklines help users understand revenue trends over time. The five slicers improve the flexibility of the dashboard by allowing users to filter information based on year, store type, state, customer segment, and product category.

The separate KPI tooltip page further improves the dashboard by presenting Total Revenue, Total Profit, Total Orders, and Average Order Value in a compact format.

Overall, this project demonstrates how Power BI can transform raw retail sales data into meaningful business insights. It helps users understand sales performance, identify profitable product categories, compare store contributions, and support informed business decisions.